THE NOVA RESEARCH CURRICULUM

Mathematics

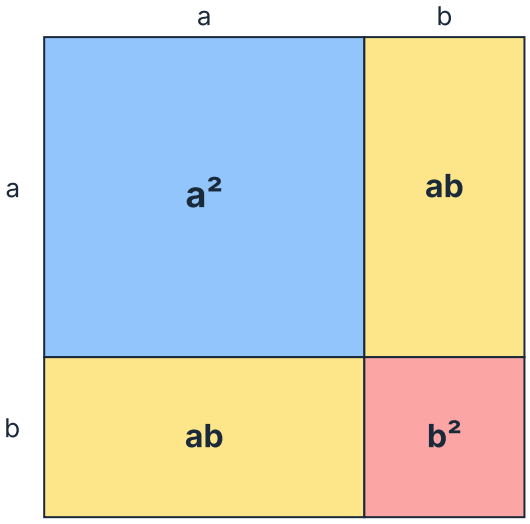
Grade 7 · Term 2

Student edition

Outcomes: MA-7.3 polynomials and identities; MA-7.4 linear equations and straight-line graphs; MA-7.5 polygons, triangle constructions and circle properties.

Lesson notes

- **MA-7.3 (Weeks 1–4):** add and subtract polynomials by collecting like terms; multiply $(x + 2)(x + 3) = x^2 + 5x + 6$. Identities: $(a + b)^2 = a^2 + 2ab + b^2$; $(a - b)^2 = a^2 - 2ab + b^2$; $a^2 - b^2 = (a + b)(a - b)$. Factorise by common factor and by identity.



$$(a + b)^2 = a^2 + 2ab + b^2$$

The square of side $(a + b)$ splits into a^2, two ab rectangles and b^2.

- **MA-7.4 (Weeks 5–8):** solve equations with brackets and unknowns on both sides. Cartesian plane: x and y axes, plotting points; table of values for $y = 2x + 1$ and draw the line.

  ▶ **Try it on NovaSchool:** Line plotter: $y = ax + b$ — open **Simulations** on your dashboard.

- **MA-7.5 (Weeks 9–12):** sum of interior angles of a polygon $= (n - 2) \times 180°$; exterior angles add to 360°. Construct triangles from SSS, SAS and ASA with ruler, compass and protractor. Circle parts: centre, radius, diameter, chord, arc, sector; the diameter is twice the radius.

Workbook pages

Page	Task
1–2	Expand and factorise 16 expressions.
3–4	Solve 10 equations; plot three lines.
5–6	Polygon angles; construct three triangles; label circle parts.

Question bank

#	Outcome	Question	Your answer
1	MA-7.3	$(x + 2)(x + 3) = ?$	$x^2 + 6$ / $x^2 + 5x + 6$ / $x^2 + 6x + 5$ / $2x + 5$
2	MA-7.3	$(a + b)^2 = ?$	$a^2 + b^2$ / $a^2 + 2ab + b^2$ / $2a + 2b$ / $a^2 - b^2$
3	MA-7.3	Factorise $x^2 - 25$.	$(x - 5)^2$ / $(x + 5)(x - 5)$ / $x(x - 25)$ / $(x + 25)(x - 1)$
4	MA-7.4	Solve $3(x - 2) = 12$.	$x = 2$ / $x = 6$ / $x = 4$ / $x = 14/3$
5	MA-7.4	Solve $5x + 4 = 2x + 13$.	$x = 3$ / $x = 17/7$ / $x = 9$ / $x = 4$
6	MA-7.4	For $y = 2x + 1$, what is y when $x = 3$?	5 / 6 / 7 / 9
7	MA-7.4	The point (0, 4) lies on the y-axis. True or false.	True / False
8	MA-7.5	Sum of interior angles of a hexagon?	540° / 720° / 360° / 1,080°
9	MA-7.5	Exterior angles of any polygon add to …	180° / 360° / 540° / it depends
10	MA-7.5	A circle has radius 7 cm. Its diameter is …	3.5 cm / 14 cm / 49 cm / 21 cm